

Riemann–Stieltjes Integration and Expectation in Statistics

1 Introduction

The *Riemann–Stieltjes integral* extends the familiar Riemann integral by integrating a function $f(x)$ with respect to another function $g(x)$. In fact, the Riemann–Stieltjes integral $\int_a^b f(x) dg(x)$ is defined via Riemann sums of the form

$$\sum_{i=1}^n f(c_i) [g(x_i) - g(x_{i-1})],$$

and reduces to the usual Riemann integral when $g(x) = x$. Thus it generalizes the Riemann integral. This added generality is useful in probability and statistics: for instance, it provides a unified framework for expectations of both discrete and continuous distributions.

2 Definition of the Riemann–Stieltjes Integral

Definition 2.1 (Riemann–Stieltjes Integral). *Let f and g be real-valued functions on the interval $[a, b]$. A partition P of $[a, b]$ is a finite sequence $a = x_0 < x_1 < \cdots < x_n = b$. For each subinterval $[x_{i-1}, x_i]$, pick a sample point $c_i \in [x_{i-1}, x_i]$. Then the Riemann–Stieltjes sum is*

$$S(P, f, g) = \sum_{i=1}^n f(c_i) [g(x_i) - g(x_{i-1})].$$

If, as the mesh of the partition goes to zero, the sums $S(P, f, g)$ converge to a limit A that is independent of the choice of points c_i , this limit is the Riemann–Stieltjes integral of f with respect to g , denoted

$$\int_a^b f(x) dg(x) = \lim_{\|P\| \rightarrow 0} S(P, f, g).$$

Typically one assumes g is a monotone (or more generally bounded-variation) function, which ensures existence of the integral in many cases.

By construction, if $g(x) = x$, the Riemann–Stieltjes integral reduces to the ordinary Riemann integral. More generally, if g is differentiable with derivative $g'(x)$, then

$$\int_a^b f(x) dg(x) = \int_a^b f(x) g'(x) dx,$$

when g' is Riemann-integrable. This shows that Riemann–Stieltjes can be thought of as a change-of-variables in integration. However, crucially, the definition does not require g to be differentiable or even continuous. For example, if g has a jump discontinuity at a point s , then $\int_a^b f(x) dg(x)$ picks up the value $f(s)$ times the jump in g at s (assuming f is continuous at s). In the extreme case of a single jump of size 1 at s , one obtains $\int_a^b f dg = f(s)$. This observation underlies how discrete distributions are handled via Stieltjes integrals in probability.

3 Properties

The Riemann–Stieltjes integral shares many properties with the Riemann integral. It is linear in the integrand: for constants α, β ,

$$\int_a^b (\alpha f(x) + \beta h(x)) dg(x) = \alpha \int_a^b f(x) dg(x) + \beta \int_a^b h(x) dg(x).$$

If g is nondecreasing, then $\int_a^b f dg$ exists whenever f is continuous (or more generally when f has at most jump discontinuities distinct from those of g). A simple sufficient condition is: if f is continuous and g is of bounded variation on $[a, b]$, then the Riemann–Stieltjes integral exists.

Theorem 3.1 (Integration by Parts). *If f and g are such that the Riemann–Stieltjes integrals below exist, then the following integration-by-parts formula holds:*

$$\int_a^b f(x) dg(x) = f(b)g(b) - f(a)g(a) - \int_a^b g(x) df(x).$$

This generalizes the usual integration-by-parts. For example, if g is differentiable, then $df(x) = f'(x)dx$ and one recovers the standard formula $\int f dg = fg - \int g df$. Integration by parts is often useful in probability, for instance in relating expectation and distribution functions.

4 Stieltjes Integration in Statistics

In probability theory, the Riemann–Stieltjes integral provides a unified definition of expectation for any real-valued random variable (discrete, continuous, or mixed). Let X be a random variable with cumulative distribution function (CDF) $F(x) = P(X \leq x)$. If f is any function (e.g. $f(x) = x$ for expectation), and the integral below exists, then the expected value of $f(X)$ is defined by the Stieltjes integral with respect to F :

$$\mathbb{E}[f(X)] = \int_{-\infty}^{\infty} f(x) dF(x).$$

This definition works regardless of whether X has a probability density function. Indeed, if F is differentiable with density $p(x) = F'(x)$, this reduces to the familiar integral $\int f(x)p(x)dx$. But if X is discrete or has jumps, the Stieltjes form still applies. In particular, for any random variable X with CDF F , the identity

$$\mathbb{E}[f(X)] = \int_{-\infty}^{\infty} f(x) dF(x)$$

holds in full generality. For example, the n th moment $\mathbb{E}[X^n] = \int_{-\infty}^{\infty} x^n dF(x)$ always makes sense if F is any CDF.

5 Examples of Expected Values via Stieltjes Integrals

Example 5.1 (Discrete Distribution). *Suppose X is a discrete random variable taking values $x_1, \dots, x_k$ with probabilities $p_1, \dots, p_k$. Its CDF $F(x)$ is a step function with jumps of size p_i at x_i . By the step-function property, the Stieltjes integral for the expectation becomes a sum of jumps:*

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x dF(x) = \sum_{i=1}^k x_i (F(x_i) - F(x_i^-)) = \sum_{i=1}^k x_i p_i,$$

since $F(x_i) - F(x_i^-) = p_i$. This recovers the usual formula for the mean of a discrete distribution. For instance, if $P(X = 0) = 1/2$ and $P(X = 1) = 1/2$, then $F(x)$ jumps by $1/2$ at 0 and by $1/2$ at 1 , and the Stieltjes integral gives $\mathbb{E}[X] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}$.

Example 5.2 (Continuous Distribution). *Let X be uniformly distributed on $[0, 1]$. Then $F(x) = x$ for $0 \leq x \leq 1$ (and 0 for $x < 0$, 1 for $x > 1$). We compute*

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x dF(x) = \int_0^1 x d(x) = \int_0^1 x dx = \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}.$$

Here we used the fact that $dF(x) = dx$ on $[0, 1]$ because $F(x) = x$ has derivative 1. This agrees with the well-known $\frac{1}{2}$ mean of the uniform distribution. Similarly, if F were a smooth CDF with density $f(x) = F'(x)$, one would write $\int x dF = \int x f(x) dx$.

6 Integration by Parts for Stieltjes Integrals

As noted above, an important property is the integration-by-parts formula. In practice, one sometimes uses it to simplify Stieltjes integrals. For example, if F is a CDF and $u(x)$ is a differentiable function, one can write

$$\int_a^b u(x) dF(x) = [u(x)F(x)]_a^b - \int_a^b F(x) du(x).$$

If F is continuous at the endpoints, the boundary terms simplify, and one often obtains alternative expressions for expectations or covariances. The general theory guarantees this formula whenever the integrals involved exist.

Summary. The Riemann–Stieltjes integral $\int f dg$ generalizes the usual integral by allowing a nontrivial integrator function g . It is formally defined as a limit of sums $f(c_i)[g(x_i) - g(x_{i-1})]$. When used in probability, taking $g(x) = F(x)$, the CDF of a random variable, one gets the unified expression $\mathbb{E}[f(X)] = \int f(x) dF(x)$ that handles discrete and continuous cases together. Integration by parts and other properties carry over, making the Riemann–Stieltjes integral a powerful tool in statistics.